SOFTWARE ENGINEERING LAB 2: Agile Backlog Creation & Sprint Simulation in Jira

Name: Sharanya C Bharadwaj

SRN: PES2UG24CS458

Section: 4H

Case Study Title: EV Charging Station Reservation System

EPICS:

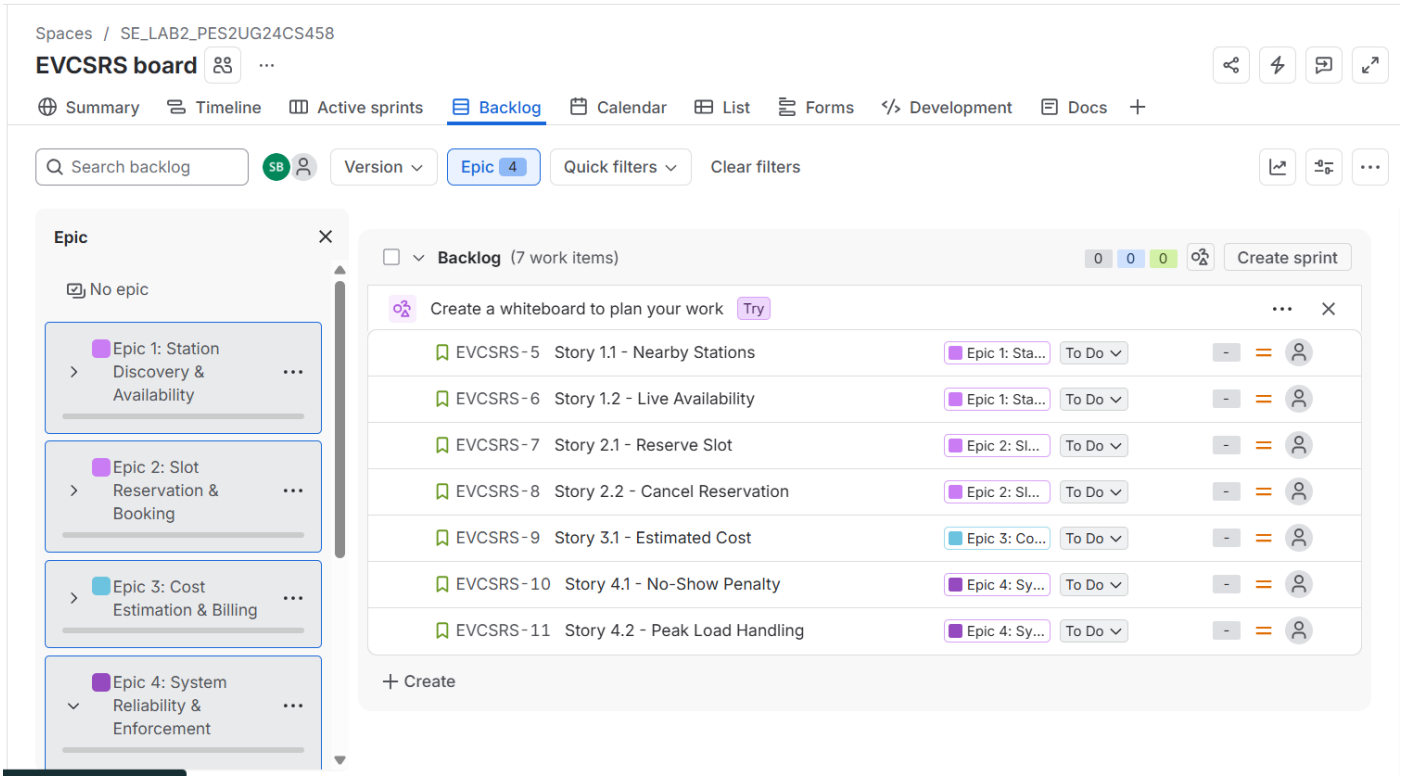
Epic 1: Station Discovery & Availability			
Description: Enable drivers to find nearby compatible stations and see real-time socket availability.			
User Stories:	As a	I want to	So that
Story 1.1: Nearby Stations	As a driver	I want to see all compatible charging stations within 10 km sorted by distance	So that I can quickly find a reachable station
Story 1.2: Live Availability	As a driver	I want to see real-time socket status (available/occupied/reserved)	So that I don't travel to a station that's actually full

Epic 2: Slot Reservation & Booking			
Description: Let drivers reserve and manage a charging slot at a compatible socket.			
User Stories:	As a	I want to	So that
Story 2.1: Reserve Slot	As a driver	I want to reserve a 30-minute charging window at a compatible socket	So that my car isn't double-booked with another driver
Story 2.2: Cancel Reservation	As a driver	I want to cancel my reservation up to 5 minutes before it starts	So that I have flexibility and free up the charger for others

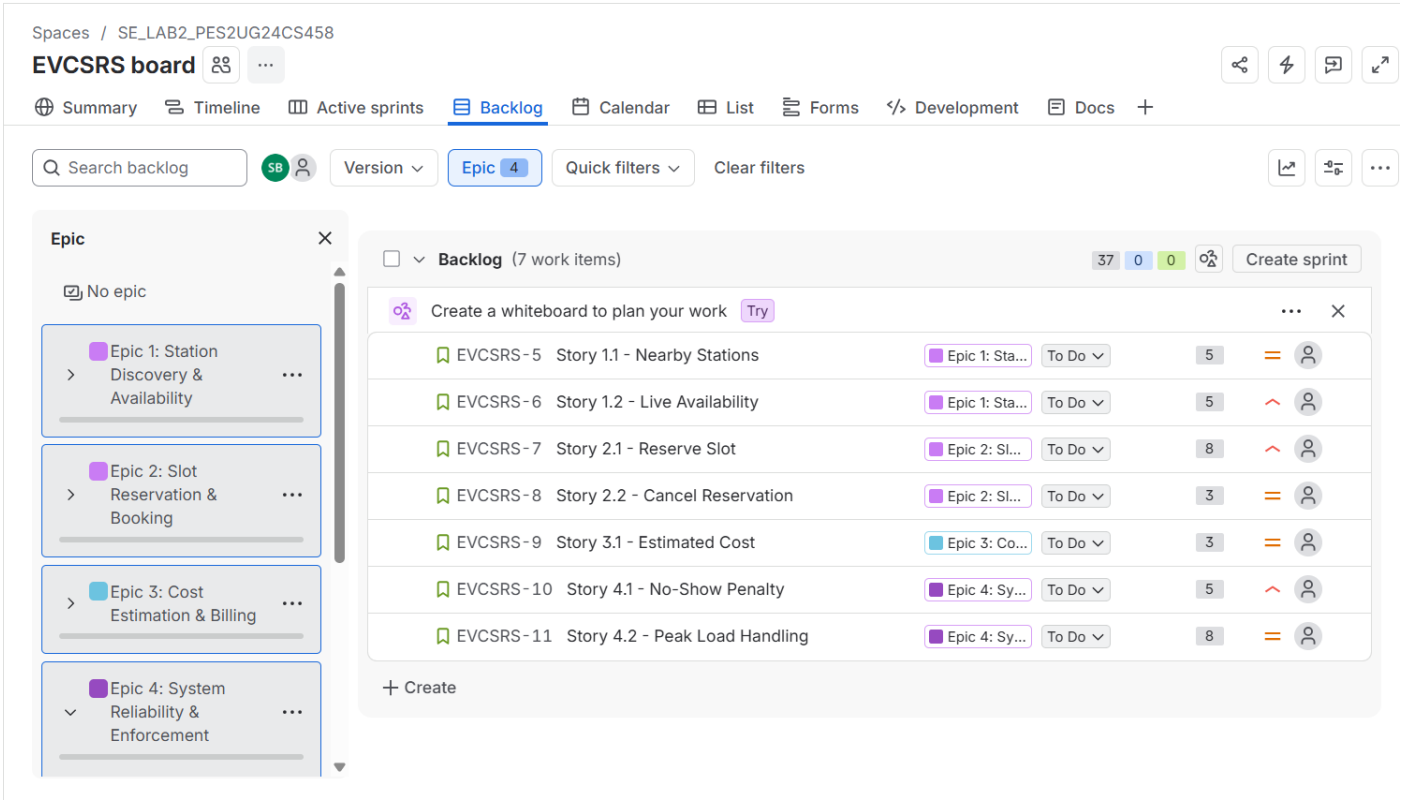
Epic 3: Cost Estimation & Billing			
Description: Show drivers an accurate cost estimate before they commit to a reservation.			
User Stories:	As a	I want to	So that
Story 3.1: Estimated Cost	As a driver	I want to see the estimated charging cost based on the dynamic rate before confirming	So that there are no billing surprises

Epic 4: System Reliability & Enforcement			
Description: Keep the system fast and prevent socket hoarding.			
User Stories:	As a	I want to	So that
Story 4.1: No-Show Penalty	As a system	I want to auto-release a locked socket and apply a penalty if the driver doesn't plug in within 10 minutes	So that chargers don't sit idle
Story 4.2: Peak Load Handling	As a driver	I want the system to respond within 3 seconds even with 500 concurrent users	So that booking doesn't lag during peak commute hours

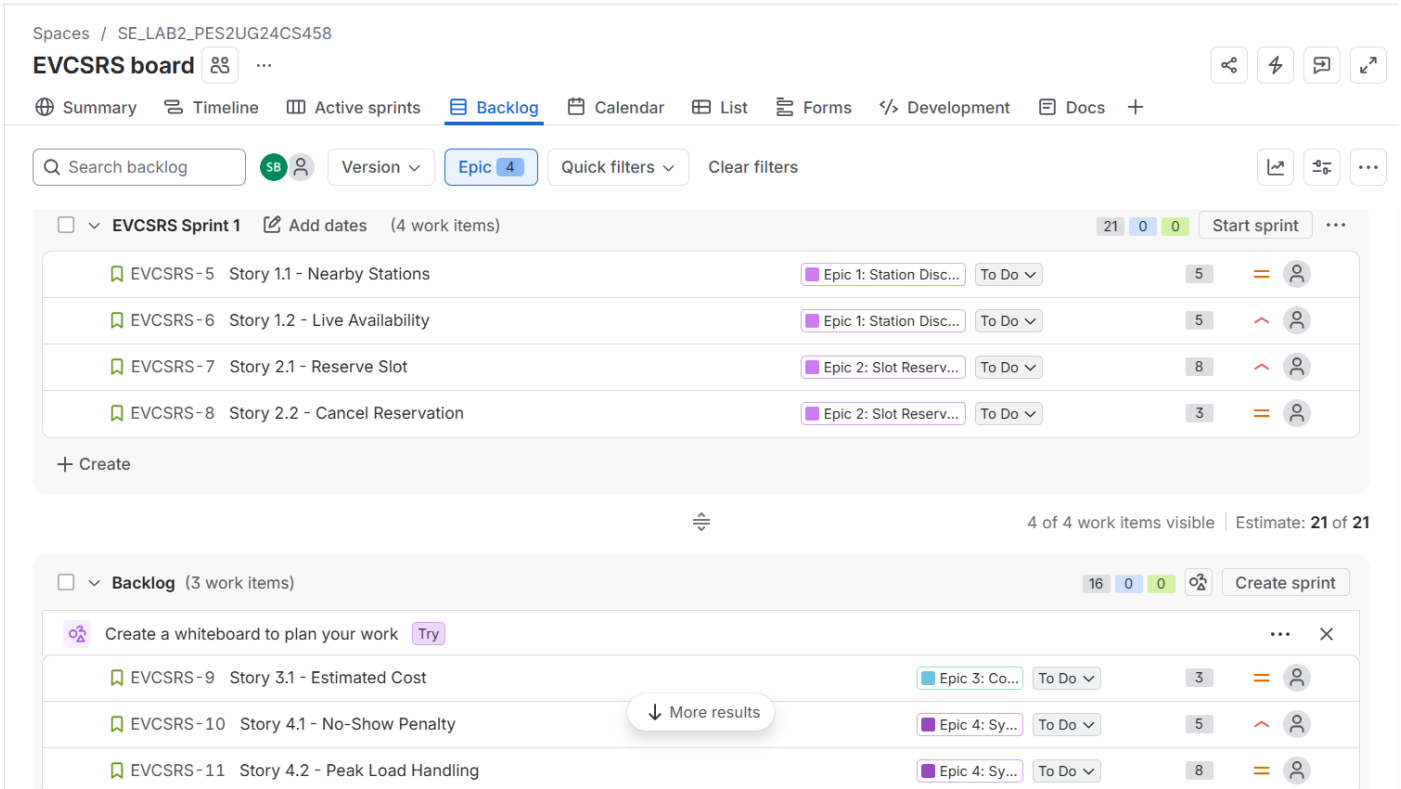
SCREENSHOTS:



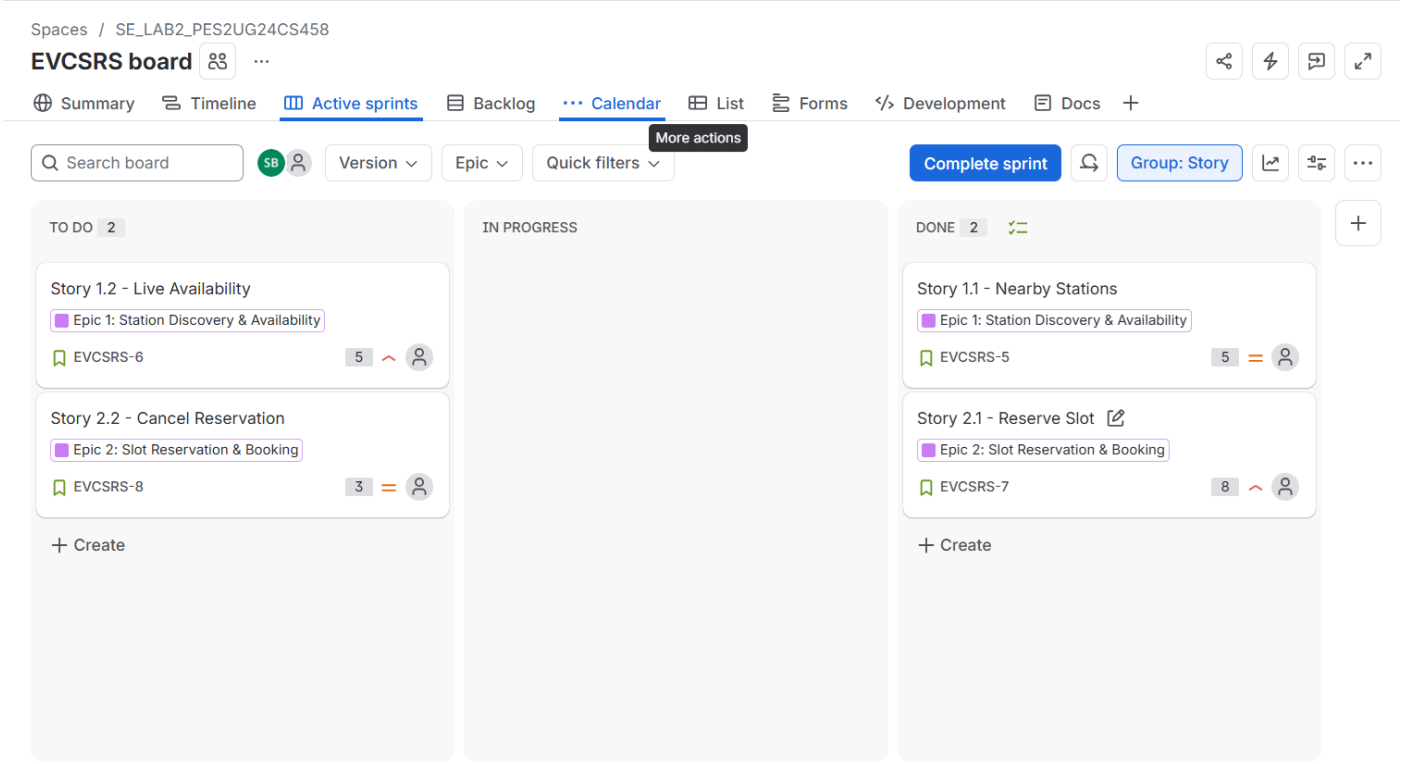
Backlog filtered by Epic, showing all 4 epics in the side panel with their linked stories.



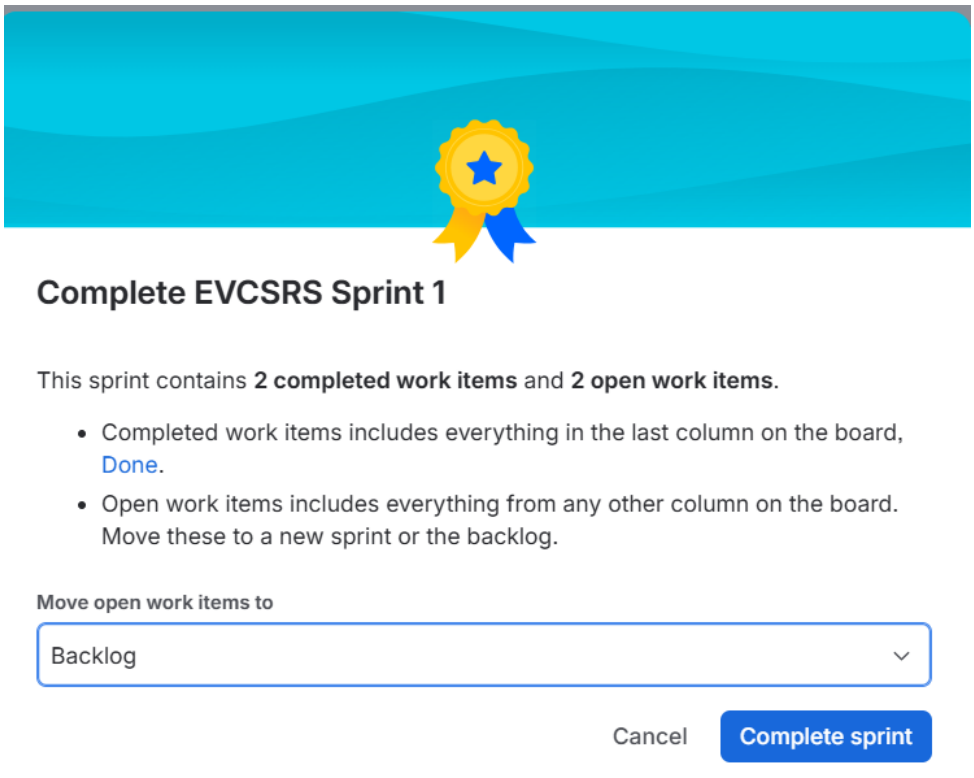
Backlog with story points and priorities assigned to all 7 stories (37 total points).



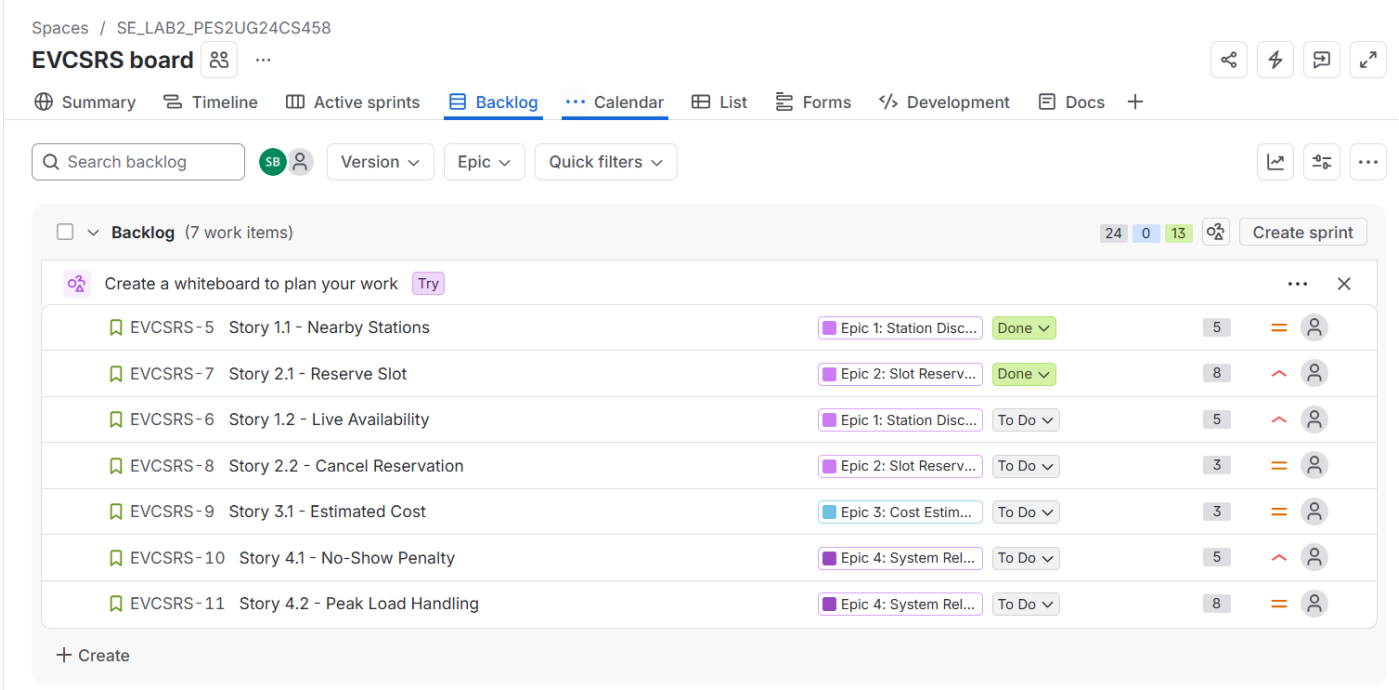
Sprint 1 created with 4 stories (21 points), remaining 3 stories left in Backlog.



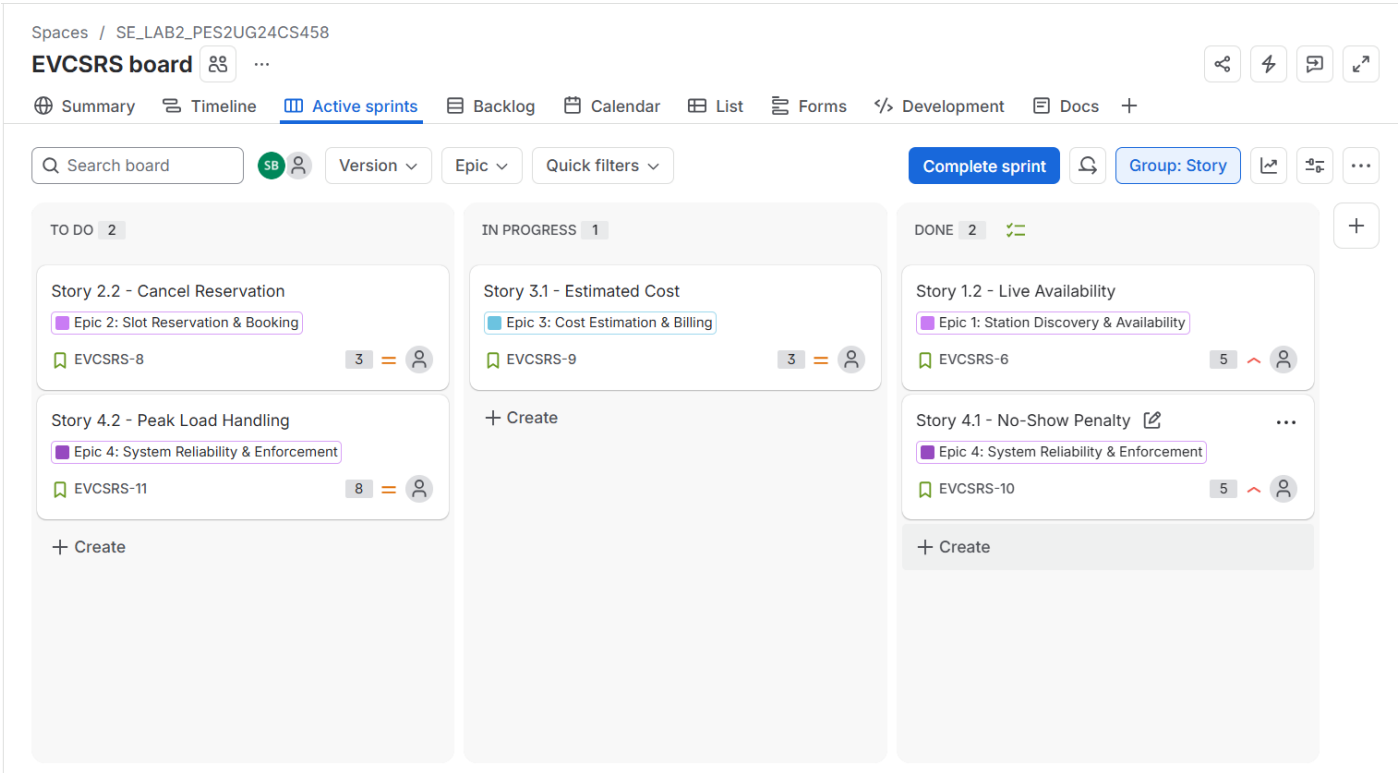
Active Sprint 1 board mid-progress - 2 stories in To Do, 2 in Done.



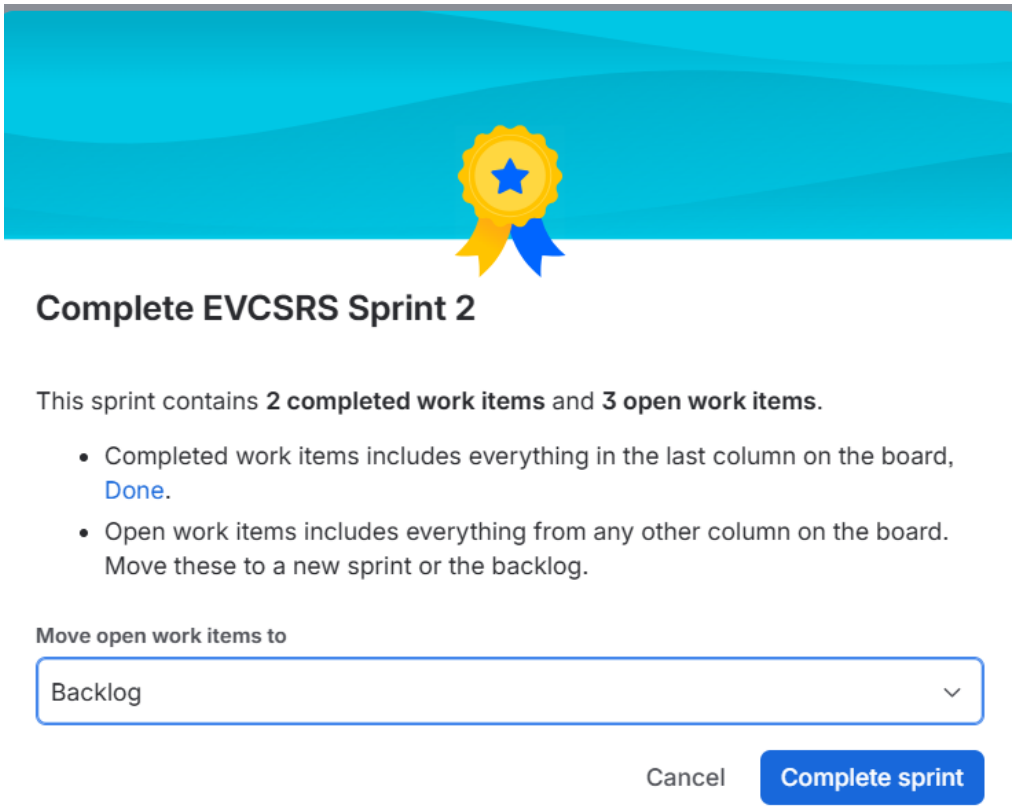
Completing Sprint 1 - 2 items completed, 2 open items moved back to Backlog.



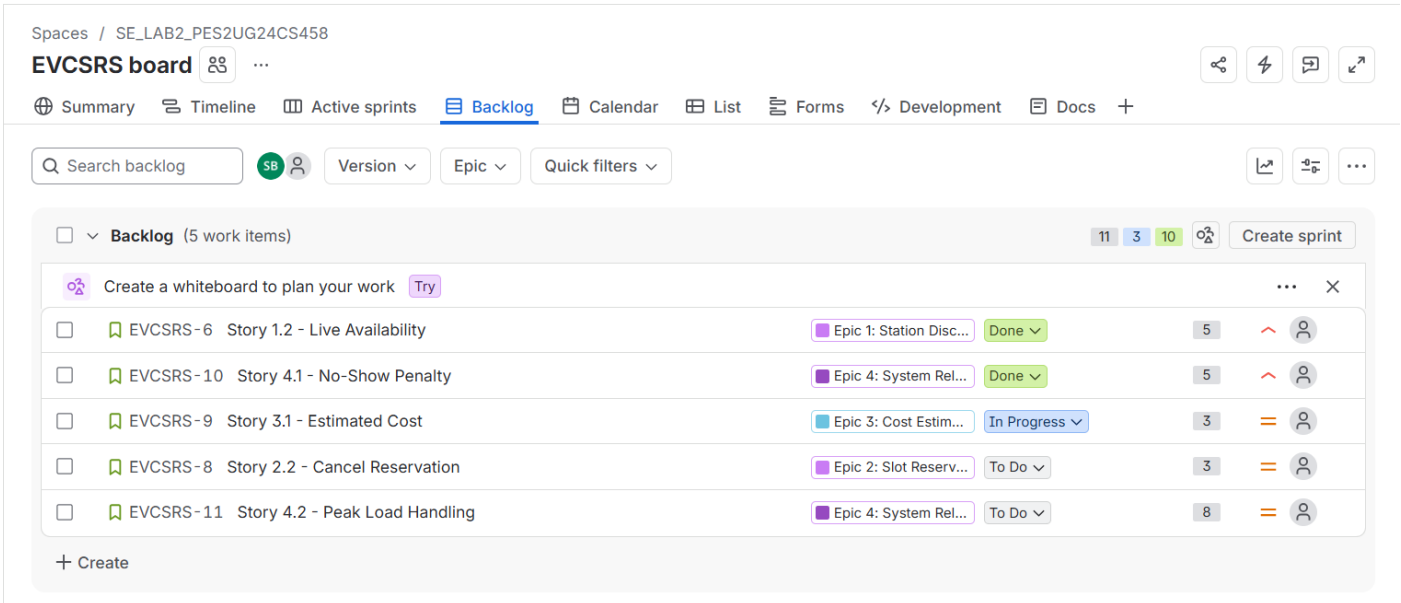
Backlog after Sprint 1 completion, showing 2 stories marked Done and unfinished ones back in To Do.



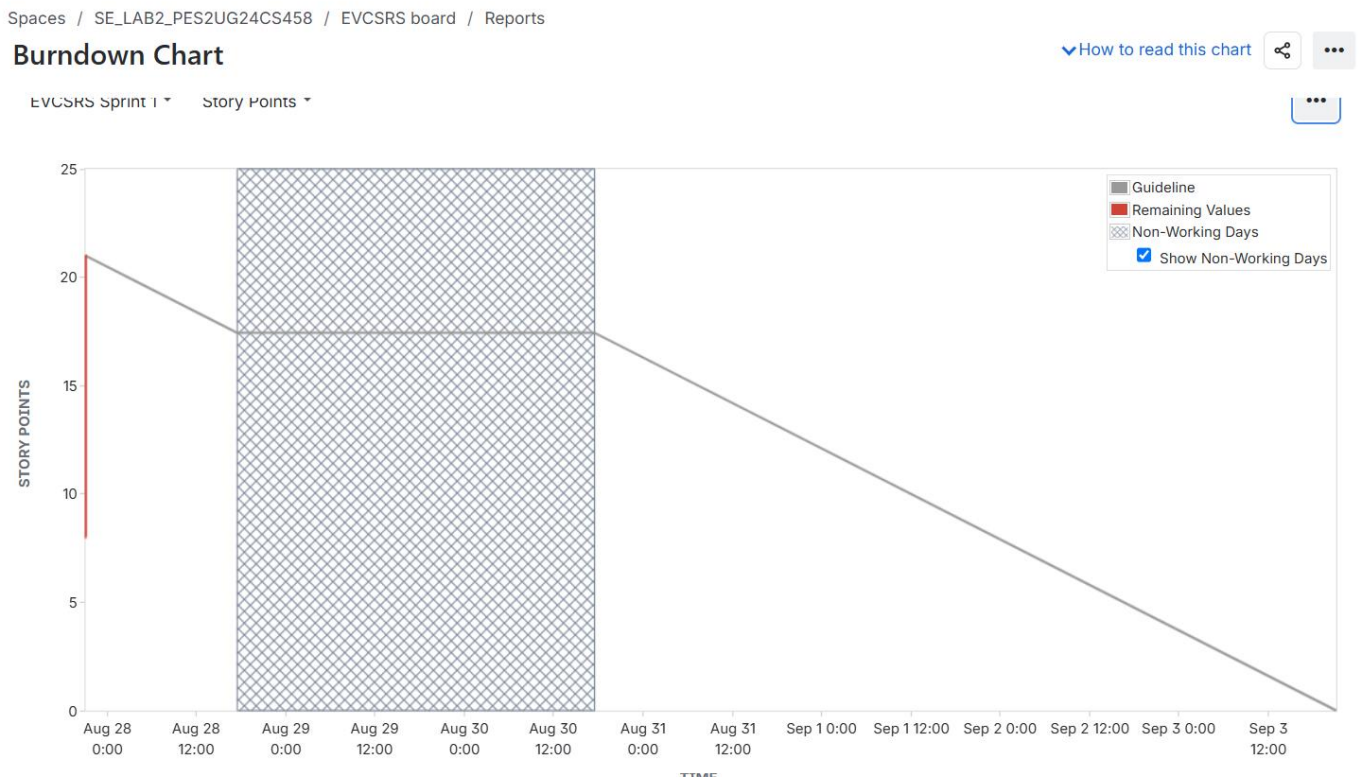
Sprint 2 board mid-progress - 2 in To Do, 1 In Progress, 2 in Done.



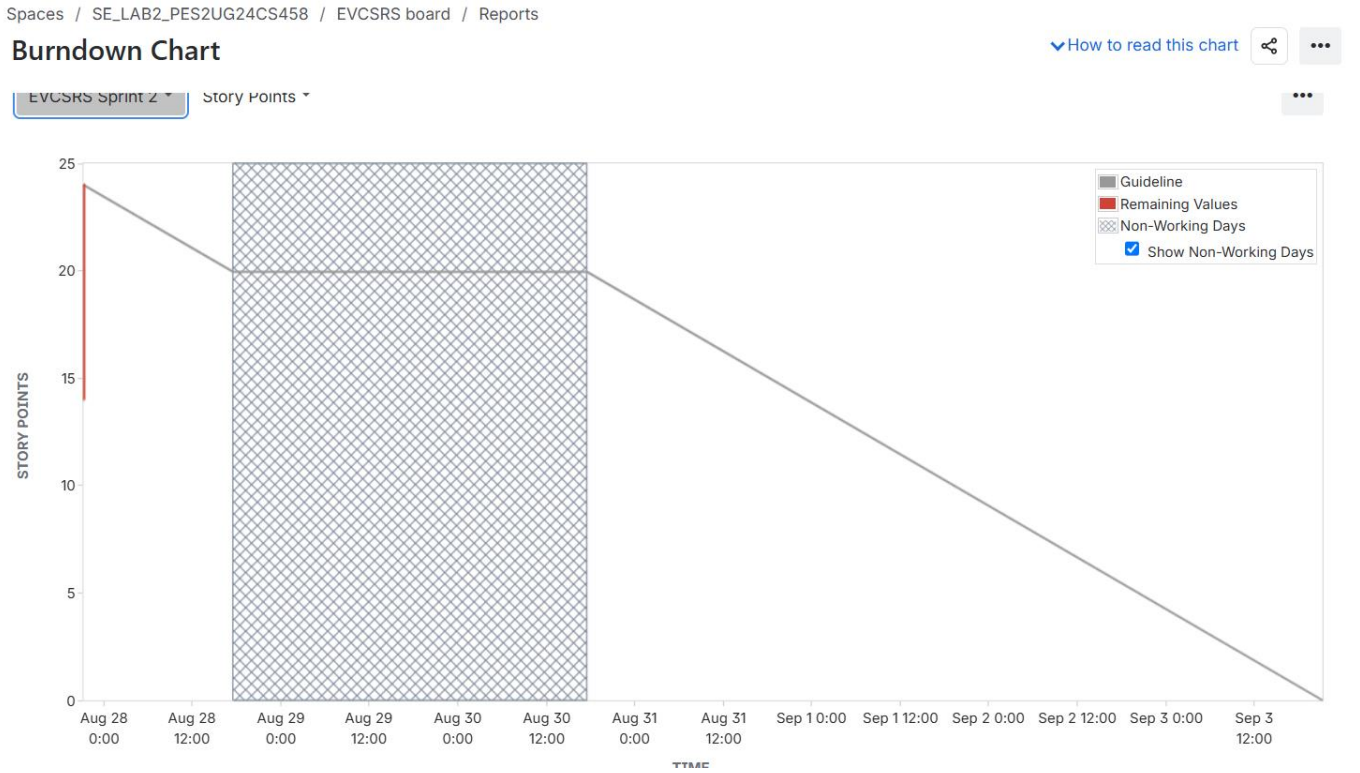
Completing Sprint 2 - 2 items completed, 3 open items moved back to Backlog.



Backlog after Sprint 2 completion, showing updated statuses across remaining stories.



Backlog after Sprint 2 completion, showing updated statuses across remaining stories.



Burndown Chart for Sprint 2, showing a similar sharp drop in remaining story points.

Reflection Questions:

1.Did your estimations reflect the actual effort?

Ans: Not really, points were assigned by guesswork upfront, and completion in both sprints happened all at once (not gradually), so there's no real effort data to compare against.

2.Was your backlog well-prioritized?

Ans: Partly , High-priority stories (1.2, 2.1, 4.1) got picked up across the two sprints, but some (like 1.2 and 4.2) still slipped to later sprints instead of finishing first.

3. How did your simulated sprint align with your plan?

Ans: Not fully , Sprint 1 only completed 2 of 4 stories (13/21 pts), and Sprint 2 only completed 2 of 5 (10/24 pts), so more than half the committed work was pending each time.

4. What insights did the burndown chart give about your team’s capacity?

Ans: The red "remaining values" line drops sharply right at the start instead of tracking the grey guideline, showing work was marked Done in a burst rather than steadily, and that both sprints over-committed relative to what actually got closed out.

Github Repository Link:

https://github.com/sharanyacbharadwaj/SE_LAB_PES2UG24CS458_SEC_H